

Hyperledger Fabric Adapter — Operator Manual

What this technology does

The Fabric adapter consumes **eligible** rows from `telemetry.fabric_outbox`, verifies/seals the local evidence boundary through OpenBao, prepares a source-bound HRC Fabric transaction, submits it through the private Fabric Gateway, and reconciles the authoritative commit/query/digest result.

It is asynchronous by design. Fabric or OpenBao downtime must not make ordinary sensor telemetry disappear.

Current production boundary

```

ULC-01 -> fabric-adapter-1 -> ENABLED production writer
ULC-02 -> fabric-adapter-2 -> installed / healthy / write-disabled standby
ULC-03 -> no Fabric adapter

Gateway endpoint: 10.104.0.7:7051
TLS server name:  peer1.hrc.local
MSP:              HrcMSP
Channel:          hrc-channel
Chaincode:        hrc-evidence
Source namespace: lorawan-gateway-evidence

```

Do not enable ULC-02 merely because ULC-01 is temporarily unreachable. Its ownership/fencing takeover remains a separate HA acceptance boundary.

Fresh live checkpoint — 2026-09-18

Read-only restricted monitoring proves ULC-01 is `enabled=true / status=ready` and ULC-02 is `enabled=false / status=standby`; both canonical self-tests pass. The same snapshot reports 2094 Fabric outbox rows and all 2094 are confirmed, with 2094 verifier results all verified. The old pending backlog is not current.

Do **not** run takeover yet: ULC-01 reports image `gateway-fabric-adapter:fairness-20260915`, while ULC-02 reports `9e08e25fe5dd`. Until exact current image/binary parity, distinct worker identity, live migration-003 fencing, and the controlled takeover row are proven, ULC-02 stays disabled.

Current HRC contract

```

CreateSourceBoundAnchor(SourceRecordID, sourceType, producer, producedAt, schemaVersion)
  transient hrc.exact_payload = immutable finalized_payload bytes

QuerySourceBoundAnchor(SourceRecordID)

VerifySourceBoundDigest(SourceRecordID, sha256(finalized_payload))

```

The retired `CreateAnchor`, `QueryAnchor`, `QueryAnchorByRecordID`, and `VerifyDigest` APIs must not be restored.

The exact payload is always `telemetry.fabric_outbox.finalized_payload`. Never reconstruct it from JSONB, `raw_data`, maps, structs, or projections.

Eligibility rule

A fresh claim requires:

```
finalized_payload IS NOT NULL
```

For telemetry-attestation-v2, it additionally requires the matching verifier-owned:

```
gateway_evidence.event_verification.status = 'verified'
```

Therefore pending with attempts=0 can be correct while the adapter itself is healthy.

Current lifecycle rule

```
pending
processing
failed
reconciling
confirmed
needs_attention
dead_letter
```

submitted_unknown remains legacy schema compatibility and is not emitted by the current worker.

An uncertain post-submit result enters reconciling. Prepared transaction bytes and the signed commit-status request are persisted before orderer submission so recovery can continue the **same transaction** rather than creating a new one blindly.

Step 1 — Prove writer ownership first

Run on ULC-01 and ULC-02:

```
id="$(sudo docker ps -q --filter 'label=com.docker.compose.project=lorawan-gateway-evidence' --filter 'label=com.docker.compose.service=fabric-adapter' | head -n1)"

if [ -z "$id" ]; then
  echo 'FABRIC_ADAPTER_CONTAINER=ABSENT'
else
  sudo docker inspect "$id" --format 'name={{.Name}} status={{.State.Status}} restarts={{.RestartCount}} image={{.Image}}'
  sudo docker port "$id" 8080/tcp
fi
```

Expected ownership:

- ULC-01: enabled production writer;
- ULC-02: healthy standby with writes disabled.

If both are enabled, treat that as an HA-fencing incident. Do not let both continue while investigating.

Step 2 — Check /healthz and /readyz

```
id="$(sudo docker ps -q --filter 'label=com.docker.compose.project=lorawan-gateway-evidence' --filter 'label=com.docker.compose.service=fabric-adapter' | head -n1)"
```

```

endpoint="$(sudo docker port "$id" 8080/tcp | head -n1)"
host="${endpoint%:*}"
port="${endpoint##*:}"
[ "$host" = "0.0.0.0" ] && host=127.0.0.1

curl --connect-timeout 3 --max-time 5 -fsS "http://${host}:${port}/healthz"
echo
curl --connect-timeout 3 --max-time 8 -fsS "http://${host}:${port}/readyz"
echo

```

PASS means:

- ULC-01 reports alive and enabled, with readiness dependencies/self-tests satisfied;
- ULC-02 reports standby, enabled=false, and canonical self-test pass.

A healthy standby should not open production DB/OpenBao/Fabric authority merely to prove the process exists.

Step 3 — Check recent bounded logs

```

id="$(sudo docker ps -q --filter 'label=com.docker.compose.project=lorawan-gateway-evidence' --filter 'label=com.docker.compose.service=fabric-adapter' | head -n1)"

sudo docker logs --since=15m --tail=180 "$id"

```

Look for repeated DB/OpenBao/TLS/Fabric errors, lease-generation loss, preparation/persistence failure, commit-status timeout, reconciliation failure, source-bound Query mismatch, or Verify mismatch.

Do not print protected adapter env files, RoleID/SecretID contents, client private keys, or unrestricted DSNs.

Step 4 — Diagnose the outbox before blaming Fabric

Using an approved read-only SQL session:

```

SELECT status, count(*) AS rows
FROM telemetry.fabric_outbox
GROUP BY status
ORDER BY status;

SELECT
    outbox_id,
    source_event_key,
    observed_at,
    schema_version,
    status,
    attempts,
    finalized_payload IS NOT NULL AS has_finalized_payload,
    worker_id,
    lease_generation,
    lease_expires_at,
    fabric_tx_id,
    last_error_category,
    last_error
FROM telemetry.fabric_outbox
ORDER BY outbox_id DESC
LIMIT 30;

```

For a v2 row that remains pending, verify the exact matching evidence result on both source identity and observation time:

```
SELECT source_event_key, observed_at, status, verified_at
FROM gateway_evidence.event_verification
WHERE source_event_key = '<SOURCE_EVENT_KEY>'
AND observed_at = '<OBSERVED_AT>':timestampz;
```

Do not manually manufacture finalized_payload, verified, confirmed, or a replacement outbox row.

Step 5 — Verify dependency boundaries separately

The adapter depends on:

```
PostgreSQL/PgBouncer -> pgbouncer.internal.lorawan.com:6432
OpenBao              -> https://openbao-kms.internal.lorawan.com:18200
Fabric Gateway       -> 10.104.0.7:7051 / peer1.hrc.local
```

Use the PostgreSQL/PgBouncer and OpenBao manuals first.

For Fabric transport only, verify server TLS without disabling hostname checks:

```
openssl s_client -connect 10.104.0.7:7051 -servername peer1.hrc.local -verify_hostname peer1.hrc.local
-verify_return_error -CAfile /etc/lorawan-pki/fabric/tls-ca.crt </dev/null 2>/dev/null | grep -E
'Verify return code|subject=|issuer='
```

A passing TLS handshake does not prove endorsement, orderer submission, peer delivery, or commit status.

Step 6 — Separate Fabric transaction stages

Troubleshoot in this order:

1. outbox eligibility
2. adapter DB lease/generation ownership
3. OpenBao login/sign/verify
4. Fabric Gateway TLS/session
5. proposal/endorsement
6. prepared transaction persistence
7. orderer submission
8. peer commit delivery / CommitStatus
9. QuerySourceBoundAnchor field match
10. VerifySourceBoundDigest = MATCH

A stale HRC peer ledger can make client CommitStatus wait even after endorsement and orderer submission succeeded. Do not classify that as a generic “submit hang.”

Step 7 — Verify the HA fencing contract

The current generation-aware claim contract requires every worker-owned mutation to match:

```
outbox_id
worker_id
lease_generation
lease_expires_at > now()
```

A successful claim increments lease_generation. A stale process cannot regain write authority merely by reusing the same stable worker ID.

ULC-02 must remain disabled until the controlled takeover gate proves that both installed workers use the generation-aware build and only one owns external side effects.

Do not reuse qualification outbox row 1960.

Step 8 — Run the activation preflight only when activation is intended

The tracked read-only gate is:

```
sudo ./fabric-adapter-enable-preflight.sh /etc/lorawan-cloud/gateway-evidence/release.env /etc/lorawan-cloud/gateway-evidence/host.env
```

Required marker:

```
FABRIC_ADAPTER_ENABLE_PREFLIGHT=PASS
```

It verifies protected file modes, PgBouncer TLS DSN, OpenBao route, Fabric certificate/key pairing, HRC CA/TLS identity, current source-bound function names, and the exact enabled Compose model without starting or mutating anything.

Do not run the enabled Compose overlay on ULC-02 as a routine health check.

Safe adapter change procedure

1. Prove ULC-01 is the sole enabled writer.
2. Check outbox and reconciliation state first.
3. Back up the exact deployment/configuration being changed.
4. Preserve durable prepared-transaction/reconciliation fields.
5. Validate the new immutable image/build and migration compatibility.
6. Keep ULC-02 disabled unless performing the explicit HA gate.
7. Run the activation preflight on the intended owner.
8. Roll only that adapter.
9. Require /readyz.
10. Confirm one new **eligible** representative row only when a functional write test is warranted.
11. Require authoritative commit status, source-bound Query match, and Verify MATCH.

Never “recover” an uncertain submission by clearing durable transaction fields and creating a fresh transaction.

Failure behavior

Fabric unavailable

```
LoRaWAN + Node-RED + PostgreSQL telemetry continue  
eligible fabric_outbox accumulates/reconciles  
adapter retries according to governed state
```

OpenBao unavailable

```
telemetry continues
adapter cannot complete sealing/submission
outbox waits/retries
```

ULC-01 adapter lost

Keep ULC-02 fenced unless the explicit takeover procedure is being executed. Database leases alone do not authorize an unreviewed external side-effect takeover.

Troubleshooting map

Symptom	First check
pending attempts=0	finalized payload + v2 verifier eligibility
adapter alive but not ready	DB/OpenBao/Fabric dependency reported by readiness/logs
OpenBao login denied	RoleID/SecretID lifecycle/policy; do not broaden policy
proposal/endorsement fails	Fabric identity/channel/chaincode/function
submit accepted, CommitStatus waits	peer ledger/delivery state vs orderer submission
reconciling persists	same prepared tx + commit-status/query recovery path
needs_attention	preserve evidence and investigate conflict/permanent failure
duplicate/conflicting anchor	stop external writes and inspect fencing/source identity
ULC-02 writing unexpectedly	fencing/enablement incident

Completion checklist

- ULC-01 is enabled and ready;
- ULC-02 is healthy but enabled=false;
- exact HRC source-bound API names are configured;
- exact immutable finalized_payload rule is preserved;
- v2 verifier eligibility is preserved;
- DB/OpenBao/Fabric TLS dependencies are healthy;
- no unexplained reconciliation/permanent conflict is present;
- any representative submission ends in authoritative commit + Query match + Verify MATCH;
- qualification row 1960 is never rerun.

Detailed references

- [../server/integrations/hyperledger-fabric/02-fabric-network-handoff.md](#)
- [../server/cloud-production/20-openbao-and-fabric-adapter.md](#)
- [../evidence-services/cloud/deploy/README.md](#)
- [../evidence-services/cloud/packaging/FABRIC-HA-FENCING.md](#)

Supporting runbook: 02-fabric-network-handoff.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

2. Fabric Network Handoff Package

The Fabric team creates and operates the network. Use this procedure to exchange the concrete values the adapter needs and to prove them in staging. Sensitive identity material must use the organization's protected provisioning channel rather than a normal message or issue.

HRC handoff status — 2026-09-08: ULC-01 EXTERNAL QUALIFICATION PASS. hrc-evidence version 1.1 / sequence 2 is live-qualified across all three Fabric peers. The Adapter uses the Task 37 source-bound API: `CreateSourceBoundAnchor(SourceRecordID, sourceType, producer, producedAt, schemaVersion)` with exact payload bytes only in transient key `hrc.exact_payload`, then `QuerySourceBoundAnchor(SourceRecordID)` and `VerifySourceBoundDigest(SourceRecordID, observedDigest)`. The legacy `CreateAnchor`, `QueryAnchor`, `QueryAnchorByRecordID`, and `VerifyDigest` functions are forbidden for Adapter identities. The frozen source identity is `lorawan-gateway-evidence`. ULC-01 (10.104.0.2) is commissioned against the restricted TLS-preserving Gateway 10.104.0.7:7051 with TLS name `peer1.hrc.local`, MSP `HrcMSP`, channel `hrc-channel`, chaincode `hrc-evidence`, and dedicated non-admin client identity `lorawan-gateway-evidence-ulc-01`; authoritative activation-preflight SHA-256 remains `d2e7c9a59270d248ea14b4bd035c671e1a9768ddb1ec8bfa527d675c3b3cca97`. The authorized one-record qualification for outbox 1960 / `SourceRecordID fdadcf14-741e-4293-ad38-39ba89369ea7` completed successfully with exact-payload SHA-256 `827e443857118c0f922c2f03e66f950a79092c74d8d8b947d489ace43d157b51`, Fabric transaction ID `dd11910c1e7c28edd11334a03cce10d80d574692733458c9059d2775be5131b8`, and HRC record ID `hrc-record-ad1258ae5ee243c01d1e091c684df1c950589bcbcd820d736a257f8cd052bfb`. The qualification runner only marks the row `confirmed` after authoritative commit success, `QuerySourceBoundAnchor` returns matching source/record/digest/payload-length fields, and `VerifySourceBoundDigest` returns `MATCH`; final outbox state is confirmed, `attempts=1`, with prepared transaction and commit-status request retained. The one-shot qualification container is gone. As re-verified on 2026-09-09, the ULC-01 continuous Adapter is active with `FABRIC_ADAPTER_ENABLED=true`; the production activation preflight passes and 325 records have reached `confirmed` after the worker start. ULC-02 (10.104.0.4) remains `FABRIC_ADAPTER_ENABLED=false/deferred`; enabling its HA worker is a separate later boundary that requires the lease-renewal/ownership-fencing gate and must not trigger a repeat of candidate 1960.

2.1 Information the Fabric team must provide

Request the following:

Item	Required value	Why it is needed
Fabric version	Exact network and peer versions	Client compatibility and support

Item	Required value	Why it is needed
Organization name	For example, AgricultureOrg or PortOperatorOrg	MSP and policy identity
MSP ID	HrcMSP	Transaction signing and authorization
Channel name	hrc-channel	Ledger routing
Chaincode name	hrc-evidence	Contract lookup
Chaincode version and sequence	Current definition (still request for change coordination)	Upgrade coordination
Contract namespace	empty/default; not Caliper alias hrcEvidence	Fabric Contract API lookup
Required transaction	CreateSourceBoundAnchor(SourceRecordID, sourceType, producer, producedAt, schemaVersion) plus transient hrc.exact_payload	Task 37 application contract
Query transaction	QuerySourceBoundAnchor(SourceRecordID)	Verification
Digest verification transaction	VerifySourceBoundDigest(SourceRecordID, observedDigest) and require MATCH	Post-commit verification
Endorsement policy	Organizations required to endorse	Retry and availability planning
Fabric Gateway endpoint	Host and port	Client connection
Peer endpoint	If the client profile requires it	Gateway and TLS routing
TLS CA certificate	Out-of-band trusted certificate	Prevent endpoint impersonation
Client identity certificate	Public certificate for the adapter	Authentication
Client private key	Securely provisioned, never emailed in plain text	Transaction signing
Connection profile	Network connection details	Client configuration
Event behavior	Commit and chaincode event format	Downstream processing
Rate limits	Transactions per second and burst limits	Back-pressure design
Maintenance windows	Expected outages and upgrade process	Operational planning

Fabric channels define membership and ledger separation, while policies determine who can act. Do not invent channel names, MSP IDs, or endorsement rules in the Node-RED flow.

2.2 Information your team must provide

Give the Fabric team:

- Gateway EUI: <GATEWAY_EUI> obtained from the active Concentrator configuration;
- LoRaWAN region: <CONFIRMED_REGION_ID> with the site authorization evidence;
- ChirpStack application and device identifier conventions;
- DevEUI normalization rule;
- the frozen EMU-01 payload-v2 decoded field dictionary, including units and sensor_validity_bitmap semantics;

- MQTT topic pattern: application/+device/+event/up;
- TimescaleDB table and view names;
- timestamp policy: UTC;
- duplicate key policy;
- expected event volume and aggregation window;
- example sanitized uplink JSON;
- example canonical attestation JSON;
- desired query and verification operations;
- data classification for each field;
- AuthenticatedSourceSystemID: lorawan-gateway-evidence;
- DigitalOcean VPC: 10.104.0.0/20;
- ULC-01 private/public source addresses: 10.104.0.2 / 143.198.205.54;
- ULC-02 private/public source addresses: 10.104.0.4 / 165.22.253.127.

Do not send passwords, private keys, LoRaWAN root keys, connection profiles containing secrets, or raw personal data in a normal project chat or issue. Exchange certificates and private keys through the approved secure provisioning channel and record only their identifiers, owners, expiry dates, and storage locations in the handoff.

2.3 Build the handoff request

Use the template below to collect the environment-specific values. Each field is consumed by the adapter configuration, TLS validation, contract invocation, capacity plan, or support procedure; omit fields that do not apply rather than inventing values:

```
Integration name: LoRaWAN telemetry attestation
Environment: pilot / staging / production
Submitting organization:
MSP ID:
Channel:
Chaincode:
Chaincode version:
Fabric Gateway endpoint:
TLS server name:
Required transaction:
Required query:
Required event:
Client identity name:
Endorsement policy:
Expected transaction rate:
Maximum accepted delay:
Private data requirement:
Test window:
Support contact:
Certificate rotation contact:
```

2.4 Test the handoff

The handoff is usable only when the Fabric team has supplied and the application team has verified:

- a reachable staging Gateway endpoint;
- a valid test identity;
- the trusted TLS CA material;
- the channel and chaincode names;
- a transaction that can be evaluated;

- a transaction that can be submitted;
- a documented commit-status result;
- a documented failure response for duplicate events;
- a certificate rotation procedure;
- a support and incident path.

Do not treat a file called `connection profile` as sufficient by itself. The adapter also needs a verified endpoint, server-name expectation, trusted TLS roots, dedicated client identity, protected private key, exact channel and chaincode contract, authorization policy, commit-status behavior, and revocation process.

2.5 Security boundary

The integration application should connect to a Fabric Gateway that belongs to the same organization as its client identity. Store the private key in a protected secret mechanism or file with restricted permissions. Never store it in Node-RED flow JSON, a dashboard, a Git repository, or a debug message.

References:

- [Fabric Gateway application model](#)
- [Certificates and identity management](#)
- [Fabric channels](#)

Next: [03-integration-architecture.md](#)

Supporting runbook: 20-openbao-and-fabric-adapter.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

20. OpenBao + Fabric Adapter for the Tiny HA POC

Status: ULC-01 TASK-37 EXTERNAL QUALIFICATION PASS. Three-node OpenBao, audit, telemetry.fabric_outbox, the Fabric-adapter runtime, database authentication, and the HRC Task 37 handoff are commissioned. HRC hrc-evidence v1.1 / sequence 2 is live-qualified and the frozen API is `CreateSourceBoundAnchor(SourceRecordID, sourceType, producer, producedAt, schemaVersion)` with exact payload only in transient `hrc.exact_payload`, followed by `QuerySourceBoundAnchor(SourceRecordID)` and `VerifySourceBoundDigest(SourceRecordID, observedDigest)`. On 2026-09-08 ULC-01 completed the explicitly authorized one-record qualification for outbox 1960 / SourceRecordID fdadcf14-741e-4293-ad38-39ba89369ea7. The exact 2828-byte payload SHA-256 was 827e443857118c0f922c2f03e66f950a79092c74d8d8b947d489ace43d157b51; the resulting Fabric transaction ID is dd11910c1e7c28edd11334a03cce10d80d574692733458c9059d2775be5131b8 and the stored HRC record ID is hrc-record-ad1258ae5ee243c01d1e091c684df1c950589bcbcd820d736a257f8cd052bfb. The one-shot runner reached confirmed only after durable prepared-transaction/commit-status state, authoritative commit success, `QuerySourceBoundAnchor` field matching, and `VerifySourceBoundDigest=MATCH`; the final outbox state is confirmed, attempts=1. Two defects encountered before the successful write were corrected without producing a duplicate Fabric proposal: the common outbox `finish()` helper now types nullable SQL parameters explicitly, and the expired ULC-01 OpenBao SecretID was refreshed using the current active OpenBao member (10.104.0.4) after audit evidence showed that SecretID issuance against a standby can fail persistence with read-only storage. The existing AppRole role/policy, Transit key, RoleID, Fabric identity, channel, chaincode, and source binding were not changed. As re-verified on 2026-09-09, Adapter-1 on ULC-01 is now the continuous production writer with `FABRIC_ADAPTER_ENABLED=true`; the authoritative activation preflight passes and 325 post-activation outbox records have reached confirmed on Fabric. The live image is sha256:d12e73ae24f7823730b6632fe8209e844c0d6125bd6a9c13e9d75cc330664f74 with service-binary SHA-256 a53990277d9032e60a9f0bfb52619a21e8c999a3573477dbfb3efe6ca6fb9591. The one-shot qualification container was removed; temporary SSH authorization and secret-bearing qualification staging were removed. Adapter-2 on ULC-02 remains `FABRIC_ADAPTER_ENABLED=false` and must stay deferred until the lease-renewal/ownership-fencing HA gate is proven. Do not rerun candidate 1960, use the old K3s-only 10.43.25.198:7051 endpoint, or use legacy Fabric APIs. Future AppRole credential rotation must respect the role's 24-hour SecretID TTL and issue against the current active OpenBao member.

20.1 Goal

Keep the future security/integration **shape** in the POC without creating another database server.

Our side owns:

```
lorawan_telemetry.fabric_outbox
Fabric adapter-1/2
OpenBao Transit KMS
integration credentials
submission/reconciliation logic
```

The external Fabric team owns its Fabric network, Gateway endpoint, organizations, channel, and chaincode.

20.2 Placement

```
ha-01
  OpenBao-1
  Fabric adapter-1

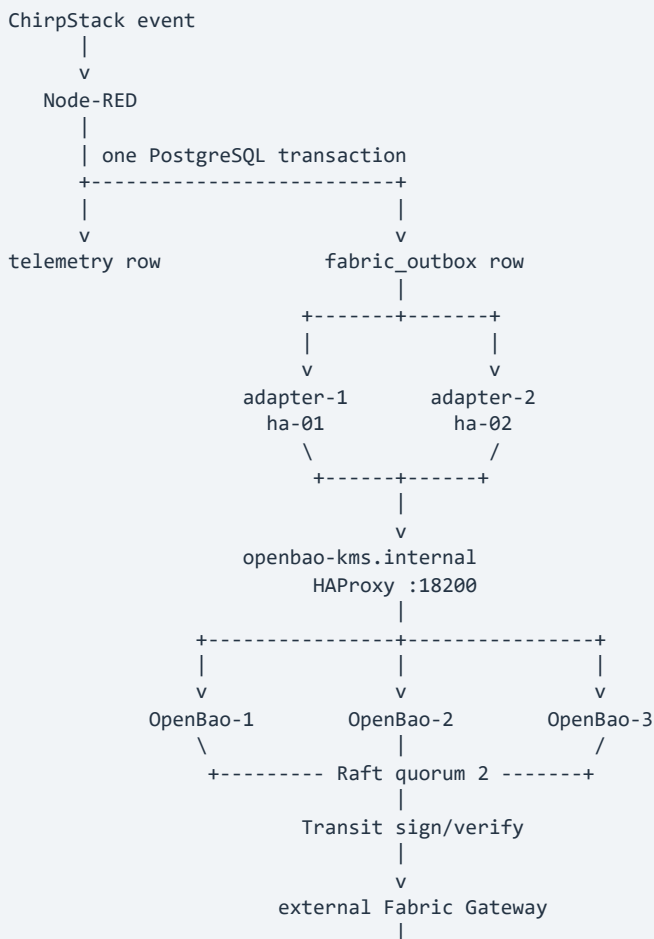
ha-02
  OpenBao-2
  Fabric adapter-2

ha-03
  OpenBao-3
```

The outbox is not on ha-03. It is a table in `lorawan_telemetry` on the **three-member Patroni PostgreSQL cluster**.

That means a PostgreSQL primary change automatically carries the outbox with the rest of the database state.

20.3 Architecture



```

      v
channel + chaincode
      |
commit status / tx ID
      |
      v
update fabric_outbox

```

20.4 Database path

Both adapters use their host-local database route:

```

Fabric adapter
-> PgBouncer :6432
-> HAProxy :15432
-> current Patroni primary :5432
-> lorawan_telemetry
-> telemetry.fabric_outbox

```

Do not give the adapters a fixed PostgreSQL node IP.

20.4A Prepared OpenBao-only parallel subphase

The OpenBao infrastructure itself has no physical-gateway dependency. While Phase 11 is compiling, the three-node KMS may be commissioned independently by following [20A. OpenBao Three-Node HA Deployment Runbook](#). This does **not** advance the Fabric adapter or full Phase 20 pass: outbox integration, adapter deployment, and one real Fabric commit still wait for their own prerequisites.

The prepared cloud pin is OpenBao 2.6.2 at OCI index digest
sha256:11fd73a2102cda9c55d5d881a8c3210303146a7ec1e8ac76f526e175c6d24641 (linux/amd64 manifest
sha256:e29524ba7c3f20d01f562c481e3eccbad6c91df45a2f2531433da4951e408cff). Normal-path setup ends after 3/3 Raft health, private TLS, Transit key/policy, HAProxy :18200, and one harmless sign/verify. Member-loss and quorum-loss remain Phase 15 tests.

20.5 OpenBao POC cluster

Use [../fabric-attestation/01-deploy-openbao-kms.md](#) as the detailed OpenBao command/config reference, but apply this **cloud POC mapping** instead of copying its larger production example blindly:

```

OpenBao node count:      3, not 5
OpenBao-1:               ha-01
OpenBao-2:               ha-02
OpenBao-3:               ha-03
API:                     private TLS :8200
Raft/cluster:             private :8201
adapter stable endpoint:  openbao-kms.internal.lorawan.com:18200 on ha-01/02 HAProxy
server SAN:               openbao-kms.internal.lorawan.com + node private name/IP

```

Bootstrap step-by-step:

1. install the same pinned OpenBao version/config pattern on all three hosts;
2. configure Integrated Storage/Raft with each node's unique node_id, api_addr, and cluster_addr;
3. install node-specific TLS key/certificate plus the common internal CA;
4. start **OpenBao-1 only** and initialize the cluster exactly once;
5. move the recovery/unseal material immediately to its protected off-node location;

6. unseal OpenBao-1 and record the initial Raft state;
7. start OpenBao-2 and join it to the existing Raft cluster—never initialize it separately;
8. unseal OpenBao-2 and verify it appears in `bao operator raft list-peers`;
9. repeat for OpenBao-3;
10. verify all three members are initialized, unsealed, and one is active;
11. enable/configure the Transit engine and create/verify the non-exportable lorawan-evidence key under the approved policy;
12. only now enable the HAProxy :18200 frontend from [07-haproxy-and-pgbouncer.md](#);
13. call `/v1/sys/health?standbyok=true` and a harmless Transit sign/verify through the stable endpoint;
14. record the healthy 3/3 Raft peer/seal state and stable-endpoint sign/verify evidence;
15. keep all three members running while proceeding to adapters.

Stop here if any node was initialized as a separate cluster, if fewer than three intended members are visible, or if HAProxy treats a sealed node as healthy. **Do not stop an OpenBao member in setup; the 2/3 survival test belongs to Phase 15.**

Use three Integrated Storage/Raft members:

```
OpenBao-1 ha-01
OpenBao-2 ha-02
OpenBao-3 ha-03
quorum = 2
```

Private ports:

```
8200 OpenBao API/TLS
8201 OpenBao Raft
18200 HAProxy stable adapter-facing KMS endpoint
```

The pre-test setup must demonstrate:

```
3/3 healthy/unsealed
Transit key exists and is non-exportable
stable :18200 endpoint is healthy
adapter identity can sign/verify through the stable endpoint
fixed canonicalization/digest vectors pass
```

The one-member-loss / 2-of-3 KMS survival experiment is reserved for Phase 15.

The evidence signing key stays non-exportable. An adapter never falls back to a local signing key.

20.6 Outbox

Create the outbox inside the Timescale-enabled `lorawan_telemetry` database on the Patroni cluster. Keep `telemetry.fabric_outbox` as an **ordinary PostgreSQL table** with the documented lease, index, permission, and immutability rules; do not convert the work queue itself into a hypertable. The telemetry event tables in the same database remain Timescale hypertables.

Required POC behavior:

```
BEGIN
  write telemetry using stable event identity
  create selected outbox row
COMMIT
```

Then the adapters process the outbox asynchronously.

Why: Fabric or KMS downtime should make the queue wait, not make sensor telemetry disappear.

20.7 Fabric writer ownership and HA target

```
ULC-01 / adapter-1 -> ENABLED production writer
ULC-02 / adapter-2 -> INSTALLED but FABRIC_ADAPTER_ENABLED=false
```

Both candidates have distinct worker identities, but lease-based row claiming alone is not sufficient proof that two processes may safely own an external Fabric side effect. The commissioned normal path therefore keeps only ULC-01 enabled. ULC-02 must remain fail-closed until the separate ownership/fencing and failover acceptance gate proves that at most one writer can submit/reconcile externally at a time.

A timeout after Fabric submission enters durable reconciliation using the already-prepared transaction and signed commit-status request; it is not permission to generate a fresh transaction. ULC-01 loss and deliberate ULC-02 takeover remain a later HA acceptance test.

20.8 External Fabric handoff

For a new environment, follow [../fabric-attestation/01-collect-external-fabric-handoff.md](#) and collect the real values from the other team. For this commissioned HRC environment, the handoff is already complete: private Gateway 10.104.0.7:7051, TLS name peer1.hrc.local, MSP HrcMSP, channel hrc-channel, chaincode hrc-evidence, empty/default contract, and source identity lorawan-gateway-evidence.

```
<FABRIC_GATEWAY_ENDPOINT>
<FABRIC_GATEWAY_PORT>
<FABRIC_TLS_SERVER_NAME>
<FABRIC_MSP_ID>
<FABRIC_CHANNEL_NAME>
<FABRIC_CHAINCODE_NAME>
submit function
query function
commit-status behavior
CA certificate
client identity
```

Do not deploy a Fabric test network on these three Droplets.

20.8A Historical server-first security readiness — commissioning complete

The adapter implementation gate below does **not** prevent independent preparation that requires no gateway hardware and no Fabric worker runtime.

The block below records the pre-activation gate that was used before Fabric credentials existed. It is retained for rebuild/recovery context and is not a statement that credentials are still absent:

```
OpenBao 3/3 healthy/unsealed through the stable :18200 endpoint
OpenBao audit-device state explicitly inspected and a reviewed audit path commissioned
fabric-adapter SecretID accessor count remains zero while no runtime exists
Node-RED selected-event atomic telemetry + outbox enqueue proven with the isolated synthetic fixture
frozen telemetry-attestation-v1 canonical-byte fixture hash reproduces exactly
expected v1 SHA-256 = c2952e8cddc7f39a17522cb49dd3292c9af75c00fdc37172f74bb3dc955f3a5c
```

The frozen v1 canonical string is already committed in ../integrations/hyperledger-fabric/04-data-contract-and-chaincode.md. Reproducing its exact SHA-256 is safe now and proves the byte fixture has not drifted. A true RFC 8785 **canonicalizer implementation** still belongs to the reviewed adapter/runtime and must pass the same vector at startup; hashing the already-frozen canonical string is not a substitute for that implementation test.

OpenBao auditing is part of the **signing** trust boundary. The Fabric security/incident-response manuals depend on KMS audit evidence to establish a possible unauthorized-signing window. The current cloud execution history does not record a commissioned audit device, so inspect that state and close the gap before releasing any signing workload credential. This does **not** block the gateway/security evidence ingest, collector, verifier, trusted-decoder, storage, or database implementation because those services hold no OpenBao/Fabric signing authority. Keep audit storage protected and separate from Raft data, use bounded host-side rotation, and avoid casually disabling/re-enabling an audit device because that destroys its existing salt/HMAC continuity.

Current commissioned state: the pre-activation gate is complete. ULC-01 is the enabled production writer and uses the commissioned least-privilege OpenBao/Fabric identity; ULC-02 retains protected standby material but remains FABRIC_ADAPTER_ENABLED=false. Do not copy one worker's SecretID or Fabric private identity to the other host, and do not enable ULC-02 until the HA ownership/fencing gate passes.

20.8B Adapter implementation readiness gate

The Fabric adapter implementation, immutable runtime image, HRC Task 37 handoff, qualification transaction, and ULC-01 continuous-writer activation are commissioned. The current recorded ULC-01 image is sha256:d12e73ae24f7823730b6632fe8209e844c0d6125bd6a9c13e9d75cc330664f74. ULC-02 remains the deliberately write-disabled standby. The remaining Fabric infrastructure gate is ULC-02 ownership/fencing failover acceptance, not another external handoff.

Before attempting adapter-1 or adapter-2, require all of these:

```
reviewed adapter source exists
build/test procedure exists
immutable image digest exists
runtime UID/GID is known
supported Fabric SDK/version is known
outbox claim/reconcile implementation matches the documented schema
OpenBao Transit client behavior is implemented
external Fabric handoff test vector passes
```

Then follow [../fabric-attestation/02-create-outbox-and-adapter.md](#), substituting the cloud endpoints:

```
lab DB host telemetry-db:5432
-> cloud pgbouncer.internal.lorawan.com:6432 / lorawan_telemetry

lab OpenBao service openbao:8200
-> cloud https://openbao-kms.internal.lorawan.com:18200

one lab adapter
-> ULC-01 adapter-1 enabled in production
-> ULC-02 adapter-2 installed with a separate protected identity but write-disabled until HA fencing acceptance
```

The adapter implementation/image gate is now **PASS**. Keep the workers in fail-closed standby until the external Fabric Gateway endpoint/TLS identity, MSP client identity, channel/chaincode/contract/function values are installed and fabric-adapter-enable-preflight.sh passes. Until that handoff is complete, do not claim a real ledger transaction or Fabric-adapter failover acceptance.

Gateway-integrity v2 has passed its implementation gate: reviewed ingest/collector/verifier/trusted-decoder components and the evidence storage path are commissioned, with real Gateway-01 lineage accepted. Keep

Fabric activation independent from that evidence lane; evidence readiness is not permission to release Fabric signing credentials.

20.9 Failure behavior

External Fabric unavailable

```
LoRaWAN          continues
Node-RED          continues on the current active host; standby remains stopped
PostgreSQL telemetry continues
fabric_outbox     accumulates
adapters         wait/reconcile/retry
```

One adapter lost

```
other worker remains
expired jobs can be reclaimed
outbox remains in PostgreSQL HA cluster
```

One OpenBao member lost

```
2/3 quorum remains
KMS endpoint remains usable
```

ha-03 lost

```
Node-RED A is lost; after fencing, Node-RED B on ha-02 is promoted
Grafana pauses
PostgreSQL stays available on ha-01/ha-02
lorawan_telemetry stays available
fabric_outbox stays available
adapter-1/2 stay alive
OpenBao remains 2/3
existing eligible outbox work can continue
```

This is a stronger POC shape than keeping the outbox in a single standalone telemetry database.

20.10 Minimal monitoring

For this POC, command-line evidence is enough:

```
OpenBao member/raft status
sealed/unsealed state
Transit sign/verify result
fabric_outbox counts by status
oldest pending age
adapter-1/2 logs
worker_id and lease owner
Fabric tx ID / commit status
```

Do not deploy a large monitoring stack solely for this integration proof.

20.11 POC backup

Before destructive tests keep:

```
lorawan_telemetry logical dump
OpenBao Raft snapshot or documented rebuild/recovery material appropriate to the test
OpenBao recovery/unseal material outside the runtime hosts
Fabric handoff metadata
adapter configuration/image reference
```

Production backup/retention is not what this POC is trying to certify.

20.12 Pre-test setup order

1. confirm Phase 12A telemetry + fabric_outbox atomic commit and retry safety
2. deploy OpenBao 3/3
3. prove Transit sign/verify through :18200 with the non-exportable key
4. run the fixed RFC 8785 canonicalization vector
5. calculate SHA-256 over the exact canonical UTF-8 bytes and verify the expected digest
6. validate the external Fabric handoff and TLS identity
7. complete the adapter implementation readiness gate

Steps 1-7 and the ULC-01 activation path below are already commissioned on the current build. For a future rebuild, re-run only the relevant gates. Do not treat this sequence as permission to enable ULC-02:

8. deploy/validate adapter-1 on ULC-01; keep adapter-2 on ULC-02 write-disabled
9. require immutable `finalized\_payload`; for v2 require verifier-owned `verified` state
10. build/verify the separate local RFC 8785/OpenBao evidence seal
11. compute SHA-256 over the exact immutable `finalized\_payload` bytes for HRC
12. prepare `CreateSourceBoundAnchor` with five metadata args + transient `hrc.exact\_payload`
13. persist txid, record_id, endorsed transaction bytes, and signed commit-status request before orderer submit
14. submit the already-prepared transaction
15. require authoritative commit status
16. require `QuerySourceBoundAnchor` field match + `VerifySourceBoundDigest=MATCH`
17. keep ULC-01/OpenBao/Fabric healthy; test ULC-02 takeover only under the explicit HA gate

Do **not** remove an OpenBao member, stop an adapter, or block Fabric connectivity here. Those are Phase 15 experiments.

20.13 Pass condition

Pre-test architecture/infrastructure pass

Require:

- outbox lives in lorawan_telemetry on the Patroni cluster;
- TimescaleDB remains inside that Patroni cluster;
- Node-RED atomically creates telemetry + selected outbox work;
- OpenBao is 3/3 healthy/unsealed through the stable endpoint;
- the Transit key is non-exportable;
- RFC 8785 canonicalization and SHA-256 fixed vectors pass;
- external Fabric handoff is complete and TLS identity is verifiable.

Full pre-test Fabric execution pass

Claim this **only when a reviewed adapter implementation/image exists** and all are proven on the healthy path:

- ULC-01 adapter-1 is the enabled writer and ULC-02 adapter-2 remains write-disabled until HA ownership/fencing is accepted;
- the selected event is converted to the fixed canonical evidence projection;
- the locally stored SHA-256 digest matches the exact canonical bytes;
- OpenBao signs/verifies the same canonical bytes and the versioned signature/key ID are persisted;
- a real selected event reaches valid external Fabric commit status;
- the returned Fabric tx ID/status is recorded in the outbox;
- read-only reconstruction can re-create the digest and verify the stored seal;
- ULC-01 remains healthy as the writer and ULC-02 remains healthy-but-write-disabled for the final recovery snapshot.

The adapter runtime itself is no longer unavailable: both immutable workers are deployed and internally commissioned. Real Fabric execution remains **externally gated** until the other Fabric system supplies its Gateway/TLS/MSP/channel/chaincode/function handoff and a real transaction can be committed and queried. Do not substitute a mock ledger for that final acceptance. Adapter loss, OpenBao member loss, Fabric outage/backlog, and reconcile/drain behavior are proven later when a real Fabric target exists.

Next required checkpoint: **Phase 13B** in [13-backup-restore-and-disaster-recovery.md](#), then Phase 14 and Phase 14B.

Supporting runbook: README.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

Reproducible Cloud Evidence Deployment Bundle

This directory is the tracked and **commissioned** server-side deployment bundle for the four Go evidence/attestation services: ingest, MQTT collector, verifier, and Fabric adapter. The base profile is live on the three cloud hosts. After the governed ULC-01 qualification, ULC-01 now runs the explicit enabled overlay as the continuous Fabric writer; ULC-02 remains intentionally disabled standby until the HA fencing gate passes. Rebuilding a server should require the repository plus protected credentials/PKI and the selected durable S3-compatible service; it should not require reconstructing Compose commands from chat history.

The Fabric adapter has two deliberately separate deployment states. The base `compose.yml` runs `adapter-1/2` with `FABRIC_ADAPTER_ENABLED=false`, so they expose only local health/readiness and do not open PostgreSQL, read an OpenBao SecretID, or contact Fabric. Actual ledger submission requires the explicit `compose.fabric-adapter-enabled.yml` overlay and must pass `fabric-adapter-enable-preflight.sh` first.

Frozen HA placement

```
ulc-01 / 10.104.0.2
  ingest-1
  collector-1
  fabric-adapter-1 enabled continuous writer -> HRC Fabric through the governed private Gateway

ulc-02 / 10.104.0.4
  ingest-2
  verifier-1
  fabric-adapter-2 disabled standby -> do not enable before HA fencing acceptance

ulc-03 / 10.104.0.8
  collector-2
  verifier-2
  no Fabric adapter worker
```

Every runtime starts with the documented ceiling 192 MiB / 0.20 CPU. That is a measurement guardrail, not permission to remove HA if the 2-GiB hosts are too small.

Files

<code>compose.yml</code>	base four-service definition; adapter is hard-frozen disabled
<code>compose.collector-mtls.yml</code>	collector client-certificate mounts
<code>compose.fabric-adapter-enabled.yml</code>	explicit activation overlay; never use for standby staging
<code>release.env.example</code>	template for immutable OCI digest references
<code>release.commissioned.env</code>	exact non-secret commissioned four-image GHCR release
<code>hosts/*.env.example</code>	exact node placement + host ports + enabled-adapter mount paths
<code>env/common.env.example</code>	non-secret common DB/S3 settings
<code>env/ingest.env.example</code>	protected ingest DB/S3 credentials + TLS paths

```

env/collector-password.env.example    legacy/reference input only; production preflight rejects password mode
env/collector-mtls.env.example        protected collector DB/S3 + client-certificate paths
env/verifier.env.example              protected verifier DB/S3 settings
env/fabric-adapter.env.example        protected DB/Fabric topology settings used only after handoff
host-preflight.sh                     early read-only live host/capacity/service/port gate
preflight.sh                          final base/standby staged-config gate
fabric-adapter-enable-preflight.sh    separate no-mutation gate before enabling ledger submission

```

The Compose definition runs every container as numeric 65532:65532, read-only root filesystem, all Linux capabilities dropped, no-new-privileges, PID limit 128, bounded Docker logs, and no in-image shell/health utility. Collector/verifier/Fabric-adapter health listeners bind host loopback only. Ingest binds only the selected private host IP/backend port. The public Evidence route is now commissioned through the shared HAProxy SNI frontend at smartagri-evidence.duckdns.org:443.

Inputs that intentionally remain external

A real base deployment must supply:

1. four OCI image **digest** references built from ../packaging/;
2. the selected SeaweedFS 4.41 raw-evidence cluster from seaweedfs/, commissioned with 010 cross-rack placement and the local internal-TLS create-only S3 frontend;
3. per-role PostgreSQL login credentials, each a member only of its corresponding NOLOGIN group role;
4. per-role S3 credentials with minimum object permissions;
5. Evidence server/client PKI and the commissioned PostgreSQL/MQTT/S3 CA files;
6. dedicated read-only MQTT collector broker identities;
7. free private/loopback host ports confirmed on the live server.

Fabric activation uses the commissioned HRC Task 37 handoff. The adapter contract is CreateSourceBoundAnchor(SourceRecordID, sourceType, producer, producedAt, schemaVersion) with the exact immutable finalized_payload bytes supplied only as transient hrc.exact_payload, followed by QuerySourceBoundAnchor(SourceRecordID) and VerifySourceBoundDigest(SourceRecordID, observedDigest). The frozen LoRaWAN source identity is FABRIC_AUTHENTICATED_SOURCE_SYSTEM_ID=lorawan-gateway-evidence; the private Gateway is 10.104.0.7:7051 with TLS name peer1.hrc.local, and ULC-01 has the commissioned dedicated non-admin adapter identity. ULC-02 remains a protected write-disabled standby until its HA fencing/ownership gate is accepted. The old 10.43.25.198:7051 K3s ClusterIP and legacy CreateAnchor, QueryAnchor, QueryAnchorByRecordID, and VerifyDigest APIs must not be used.

The repository does not guess image digests, secrets, or unused evidence-service host ports. The raw-store endpoint itself is now frozen to https://evidence-objects.internal.lorawan.com:18443; each evidence container resolves that name to its local host VPC IP, while SeaweedFS's raw S3 listener remains loopback-only.

Fixed host layout

Use these directories on every server:

```

/etc/lorawan-cloud/gateway-evidence/
  release.env
  host.env
  common.env
  ingest.env          # only on ingest hosts
  collector.env        # only on collector hosts
  verifier.env         # only on verifier hosts
  fabric-adapter.env   # ulc-01 active writer / ulc-02 protected disabled standby

```

```

/etc/lorawan-pki/gateway-evidence/
  postgres-ca.crt
  s3-ca.crt
  ingest-server.crt / ingest-server.key / gateway-client-ca.crt # ingest hosts
  mqtt-ca.crt                                           # collector hosts
  collector-broker*.crt/key                             # mTLS collector mode only

# ulc-01/02 only, created only at Fabric activation time:
/etc/lorawan-cloud/fabric-adapter/
  role_id
  secret_id

/etc/lorawan-pki/fabric/
  tls-ca.crt
  client.crt
  client.key

```

Protected role env files must be `root:root 0600`. Runtime private keys are bind-mounted into a numeric non-root container; install them as `root:<gid 65532> 0440`. Public certificates/CA files must also be readable by runtime `65532:65532`: use either `root:65532 0440` or a root-owned non-writable world-readable public file such as `0444`. The preflight checks actual mode/GID readability so a `0600 root:root` CA cannot pass and then break container startup.

If GID 65532 is unused on the server, a deployment operator may create a named host group for clarity before installing keys:

```
sudo groupadd --system --gid 65532 evidence-runtime
```

Do not change an existing GID 65532 owner blindly. If that numeric GID is already allocated, stop and review the container/runtime ownership design instead of weakening key permissions.

Prepare configuration

Copy examples, then replace placeholders **outside Git**:

```

sudo install -d -m 0750 -o root -g root /etc/lorawan-cloud/gateway-evidence
sudo install -d -m 0750 -o root -g root /etc/lorawan-pki/gateway-evidence

# Example only: choose the matching host file.
sudo install -m 0644 -o root -g root release.env /etc/lorawan-cloud/gateway-evidence/release.env
sudo install -m 0644 -o root -g root host.env /etc/lorawan-cloud/gateway-evidence/host.env
sudo install -m 0644 -o root -g root common.env /etc/lorawan-cloud/gateway-evidence/common.env
sudo install -m 0600 -o root -g root ingest.env /etc/lorawan-cloud/gateway-evidence/ingest.env

```

Do not copy literal example placeholders into production.

Database credentials

`001_gateway_evidence.sql` creates three NOLOGIN group roles:

```

gateway_evidence_ingestor
gateway_evidence_collector
gateway_evidence_verifier

```

The production LOGIN identities are now frozen and live. Preserve these mappings rather than creating new ad-hoc names:

```

evidence_ingest_ulc01    -> gateway_evidence_ingestor
evidence_ingest_ulc02    -> gateway_evidence_ingestor

```

```
evidence_collector_ulc01 -> gateway_evidence_collector
evidence_collector_ulc03 -> gateway_evidence_collector
evidence_verifier_ulc02  -> gateway_evidence_verifier
evidence_verifier_ulc03  -> gateway_evidence_verifier
```

All six are SCRAM LOGIN roles with exactly one intended NOLOGIN authority membership and inherited CONNECT on `lorawan_telemetry`. Protected plaintext custody remains outside Git/Markdown under the commissioned root-only bootstrap boundary.

Every DSN uses the commissioned logical endpoint `pgbouncer.internal.lorawan.com:6432`, database `lorawan_telemetry`, hostname-verified TLS, and the mounted PostgreSQL CA. Because this internal name is not assumed to exist in public DNS, Compose maps it inside every container to the **local node private IP** (`10.104.0.2`, `.4`, or `.8`), preserving the local PgBouncer -> local HAProxy -> Patroni-primary path. The Go pool also enforces SCRAM-SHA-256 and verifies role membership/writable-primary routing on every new physical DB session.

All six PostgreSQL LOGIN roles and the three-node PgBouncer expansion are commissioned. Every PgBouncer node has the exact ten-entry/six-evidence SCRAM userlist SHA-256 `665f1592c96ca276681a454b9cbcd6ab8ab0cbfb4594b8ddd443239db58df391`, with the original four-role verifier set preserved for rollback and proven byte-identical across the expansion. Fresh strict verify-full evidence authentication passed through all three physical `:6432` endpoints and routed to the current writable Patroni primary. Do not regenerate PgBouncer userlists again unless a later credential change requires it.

S3 credentials

The production object-store backend is selected and commissioned through S9: SeaweedFS 4.41 behind the internal HAProxy TLS/create-only endpoint `https://evidence-objects.internal.lorawan.com:18443`. The locked production helper write and cross-host retained-object verification are PASS. Runtime S3 identities are already installed with the following permission split:

```
ingest:    HeadBucket + GetObject + PutObject on evidence prefix
collector: HeadBucket + GetObject + PutObject on evidence prefix
verifier:  HeadBucket + GetObject only on evidence prefix
```

The application uses conditional create-if-absent writes. Do not grant object delete/overwrite administration to these runtime credentials.

Collector broker authentication

Production collector authentication is frozen to **mTLS** because the current commissioned `ulc-01:8884` and `ulc-02:8884` gateway-facing broker listeners require client certificates and map certificate identity to the MQTT username. Do not reopen password mode merely for convenience.

```
collector-mtls.env.example -> /etc/lorawan-cloud/gateway-evidence/collector.env
EVIDENCE_COLLECTOR_AUTH_MODE=mtls
```

Issue dedicated collector `clientAuth` identities whose Mosquitto usernames receive only `read` `as923/gateway/+/#/event/#`; grant no publish or command permission. `compose.collector-mtls.yml` is supplied automatically by `preflight.sh` and must also be supplied to the later activation command. The password example remains reference material only and is rejected by production `preflight`.

Each collector replica has **two distinct client IDs**, one per fixed backend (`10.104.0.2:8884` and `10.104.0.4:8884`). Across two collectors that produces four independent persistent read-only sessions.

Fast live-host preflight

Before creating evidence credentials, applying the migration, or choosing final ports, run the same observation-only check from an **already-authorized SSH session** on each server:

```
sudo bash host-preflight.sh
```

Run it on `ulc-01`, then `ulc-02`, then `ulc-03`. It verifies current private-IP identity, memory/disk headroom, UID/GID 65532 collision risk, Docker/Compose/Buildx availability, container RSS, HAProxy/PgBouncer state, Patroni/etcd visibility, current :8884 mTLS policy where Mosquitto lives, the existing anchor :443, and dynamically suggests currently free ingest/health ports. Suggested ports are observations only; they are not reserved and should be copied into protected host configuration immediately before the final staged-config preflight.

The script deliberately does not require the repository runner to possess server private keys. Never copy a workstation SSH private key into this repository or onto a Droplet merely to automate this check.

Mandatory final staged-config preflight

After files are installed, but **before pulling or starting anything**:

```
sudo ./preflight.sh \  
  /etc/lorawan-cloud/gateway-evidence/release.env \  
  /etc/lorawan-cloud/gateway-evidence/host.env
```

It verifies:

- exact frozen node/profile placement, including adapter-1 on `ulc-01` and adapter-2 on `ulc-02` only;
- correct private IP on the host;
- all four image references are immutable `@sha256` references;
- the base Fabric adapter profile remains standby-only and accepts no DB/OpenBao/Fabric credential mounts;
- selected host ports are numeric and currently free;
- protected env/key ownership and permissions;
- required CA/certificate files are actually readable by runtime 65532:65532 and private keys are root:65532 0440;
- `pgbouncer.internal.lorawan.com` is mapped inside containers to the local node private IP;
- production backend is `s3`, never the dev filesystem;
- collector auth-mode-specific inputs;
- PostgreSQL DSNs use the logical PgBouncer endpoint + `verify-full`;
- `docker compose config --quiet` succeeds.

Expected final marker:

```
EVIDENCE_DEPLOYMENT_PREFLIGHT=PASS
```

The script does not pull images, start containers, issue credentials, alter permissions, or mutate the database.

Base/standby activation command

This boundary was commissioned on 2026-09-01 after image digests, SeaweedFS S9, the DB/PgBouncer roles, Evidence PKI, shared-443 design, and all three host preflights passed. The command remains the reproducible start/recovery path. Production collectors use mTLS, so collector hosts include the collector override:

```
sudo docker compose \
  --env-file /etc/lorawan-cloud/gateway-evidence/release.env \
  --env-file /etc/lorawan-cloud/gateway-evidence/host.env \
  -f compose.yml \
  -f compose.collector-mtls.yml \
  up -d
```

On ulc-02 there is no collector profile, so use only `compose.yml`. The Fabric adapter profile is present on ulc-01/02 but remains disabled by the base Compose definition. Start and verify one logical boundary at a time; do not bring all six evidence replicas plus two adapter standbys up blindly.

Fabric adapter activation is a separate gate

Do not issue/install an OpenBao SecretID merely because the standby container is healthy. After the external Fabric handoff is complete, install the protected adapter env/identity files, then run:

```
sudo ./fabric-adapter-enable-preflight.sh \
  /etc/lorawan-cloud/gateway-evidence/release.env \
  /etc/lorawan-cloud/gateway-evidence/host.env
```

This preflight verifies the exact adapter host/worker identity, protected file modes, `fabric_adapter` PgBouncer TLS DSN, fixed OpenBao service path, Fabric certificate/private-key match, stable OpenBao TLS/health, external Fabric Gateway server TLS, and the combined Compose configuration. It performs no pull/start/credential issuance or state mutation.

Only after that marker is `FABRIC_ADAPTER_ENABLE_PREFLIGHT=PASS` may the explicit overlay be used for the one adapter being activated:

```
sudo docker compose \
  --env-file /etc/lorawan-cloud/gateway-evidence/release.env \
  --env-file /etc/lorawan-cloud/gateway-evidence/host.env \
  -f compose.yml \
  -f compose.fabric-adapter-enabled.yml \
  up -d fabric-adapter
```

On ulc-01 include `-f compose.collector-mtls.yml` as well when managing the whole host stack. If the external Fabric Gateway requires a separate **transport mTLS** client certificate rather than server-auth TLS plus the Fabric application identity currently implemented, stop at the preflight failure and extend/rebuild the adapter; never disable TLS verification to force the connection.

Commissioned immutable release - 2026-09-01

Tracked `release.commissioned.env` and the production release files use these immutable references. The current hosts may use protected exact preloaded-image overrides with `--pull never` because the legacy host registry credential cannot read the new private GHCR packages; do not weaken package privacy as a workaround. A rebuild should either provision authorized registry-read access or preload images whose content identity has been verified against the commissioned release.

```
EVIDENCE_INGEST_IMAGE=ghcr.io/jervis-umtc/lorawan/gateway-evidence-
ingest@sha256:f2fd695e8d2505a12202b4fabd92d035c562e16d4e308ccf9436a08a36aaec1a
EVIDENCE_COLLECTOR_IMAGE=ghcr.io/jervis-umtc/lorawan/gateway-mqtt-evidence-
collector@sha256:0bcaaa127f913157596a87077235d4f0717fa1c29f08d8aca8568c3167ed2cff
EVIDENCE_VERIFIER_IMAGE=ghcr.io/jervis-umtc/lorawan/gateway-evidence-
verifier@sha256:3416496fa563154fbd7209a575acd040b735b3c4f4159902fc886e78167195c0
EVIDENCE_ADAPTER_IMAGE=ghcr.io/jervis-umtc/lorawan/gateway-fabric-
adapter@sha256:cd4308e8985d74ea7fab957a2a4adadd1831b776a8f9af2fdd6291179df83e7a
```

All three authoritative base `preflight.sh` runs passed. Current placement is `ulc-01=ingest-1+collector-1+adapter-1-enabled`, `ulc-02=ingest-2+verifier-1+adapter-2-disabled`, `ulc-03=collector-2+verifier-2`. The base Compose remains deliberately fail-closed; ULC-01 reaches the enabled state only through the governed activation overlay and `preflight`. Every runtime uses numeric `65532:65532`, read-only rootfs, `cap_drop: ALL`, `no-new-privileges`, `pids_limit: 128`, `192 MiB`, `0.20 CPU`, bounded logs, and `restart: unless-stopped`.

SeaweedFS S9, Evidence PKI, four distinct read-only collector mTLS identities/ACLs, the dual-broker collector readiness gate, replicated ingest/verifier readiness, and the private/shared-443 normal path are PASS. ULC-01 Adapter-1 is the governed continuous Fabric writer with `FABRIC_ADAPTER_ENABLED=true`; ULC-02 Adapter-2 remains healthy `FABRIC_ADAPTER_ENABLED=false` standby until the HA fencing gate passes.

What is still not claimed

The real Gateway-01 / assembled EMU-01 v2 physical evidence lineage is PASS through Concentrator, MQTT, journal/uploader, Evidence ingest, checkpoints/segments, and trusted-decoder verification. On the final 2026-09-02 shutdown check, Gateway-01 had closed and receipted segment 28, its open journal was empty, and the cloud held 492 verified v2 events with 0 pending, 0 unseen, and 0 `evidence_gap / integrity_failure` rows. Both verifier replicas run the corrected continuous-discovery build that excludes already-seen v2 keys before applying the 100-row batch limit, preventing the old first-100 starvation condition. Public ChirpStack/Evidence/MQTT normal paths are also PASS. Remaining external claims are deliberately narrower: Reserved-IP reassignment/failover authority and controlled acceptance, plus the separate ULC-02 Fabric HA fencing/dual-worker acceptance gate. Do not redo the already-passed cloud evidence commissioning without a relevant state change.

Supporting runbook: FABRIC-HA-FENCING.md

This is the original project runbook. Check the current documentation baseline and actual host state before applying historical steps or changing a live system.

Fabric Adapter HA Fencing Commissioning

Status date: 2026-09-18

This note records the generation-fencing repair and the exact deployment boundary for commissioning the second Fabric adapter worker. It is intentionally separate from older Task 37 qualification/build evidence.

Safety boundary

- Never rerun Task 37 qualification candidate/outbox 1960.
- ULC-01 remains the sole production writer until both nodes run the generation-aware binary and migration `003_fabric_adapter_ha_fencing.sql` is present.
- ULC-02 must remain `FABRIC_ADAPTER_ENABLED=false` until image parity, distinct worker IDs, database fencing, and the controlled failover gate are verified.
- The exact-payload, durable prepared-transaction, Query, and Verify semantics remain unchanged.

Fencing contract

Migration `003_fabric_adapter_ha_fencing.sql` adds `telemetry.fabric_outbox.lease_generation` `BIGINT NOT NULL DEFAULT 0`, a nonnegative check, its column comment, and the minimum `fabric_adapter` `UPDATE` grant for that column.

Every successful claim atomically increments `lease_generation`. Every worker-owned mutation must match all of:

```
outbox_id
worker_id
lease_generation
lease_expires_at > now()
```

Lease renewal is also generation-bound. The worker performs an explicit generation-aware renewal before Fabric Prepare, before SubmitPrepared, and before reconciliation resubmission. A stale process therefore cannot regain authority merely because it later uses the same stable worker ID.

The migration is additive. It can remain installed during rollback, but dual-worker operation must not mix a generation-aware worker with an older worker that does not enforce the generation predicate.

Database-backed proof

A disposable PostgreSQL 16 instance exercised the real lease timing and SQL predicates. Result: `HA_FENCE_DB_PROOF_PASS`.

Sequence:

1. worker same-worker claimed one row as generation 1;
2. a real 92-second stall allowed the 90-second lease to expire;
3. the same worker ID reclaimed the row as generation 2;
4. stale generation 1 lease renewal affected 0 rows;
5. stale generation 1 prepared-transaction persistence affected 0 rows;
6. stale generation 1 confirmation affected 0 rows;
7. current generation 2 renewal, prepared persistence, and confirmation each affected 1 row;
8. final row was confirmed|2|current-tx|2.

The disposable PostgreSQL container was removed after the proof.

Source/build acceptance

The HA changes were overlaid onto a detached clean HEAD worktree so unrelated dirty workspace files could not contaminate acceptance. Using the pinned Go 1.25.0 toolchain with `GOENV=off`, `GOTOOLCHAIN=local`, offline module/cache settings, `-mod=readonly`, `-trimpath`, and `-buildvcs=false`:

```
GO_MOD_VERIFY=PASS
TASK_GOFMT=PASS
GO_TEST_ALL=PASS
GO_BUILD_ALL=PASS
REPRODUCIBLE_BINARY=PASS
ISOLATED_RELEASE_GATE=PASS
```

Two Linux/amd64, CGO-disabled builds with the build cache cleared between them were byte-identical:

```
gateway-fabric-adapter
size: 25585501
sha256: 91873499220db54a27c49980a51c8ea4afe0acf9d0c446681cbb6d365db3db6b
```

Locked deployment image

The test-preparation deployment image is a scratch image containing only the accepted binary, running as `65532:65532` with entrypoint `/service`. The binary label matches the accepted binary SHA-256.

```
image tag: task37-gateway-fabric-adapter:ha-91873499220db54a
image id: sha256:d139b25e60bc3b4baf32ad465b59fd8f42a764516eb984d4318a307e3a2947fc
user: 65532:65532
entrypoint: ["/service"]
binary: 91873499220db54a27c49980a51c8ea4afe0acf9d0c446681cbb6d365db3db6b
```

The workstation's available legacy Docker builder injects build-history timestamps, so repeated package builds did not produce identical image IDs. Do **not** claim OCI-image reproducibility for this HA package. Binary reproducibility is proven; runtime parity is enforced by deploying the exact same locked archive to both nodes.

Locked deployment archive:

```
filename: gateway-fabric-adapter-ha-91873499220db54a.tar
sha256: 0e863742cf6bfd8701ce9c5f370d5ec14f47470a637730ebdd82c8a1f3209007
size: 12231168
```

Live preflight — 2026-09-18

Fresh read-only checks through the restricted research-recorder boundary established the current production posture without changing either adapter:

```
ULC-01 wrapper          research-recorder-server-v7
ULC-01 database role     leader
ULC-01 Fabric adapter    HTTP ready 200 / enabled=true / status=ready
ULC-01 startup worker_id fabric-adapter-ulc-01
ULC-01 canonical self-test pass
ULC-01 live image reference gateway-fabric-adapter:fairness-20260915

ULC-02 wrapper          research-recorder-server-v7
ULC-02 database role     replica
ULC-02 Fabric adapter    HTTP ready 200 / enabled=false / status=standby
ULC-02 canonical self-test pass
ULC-02 live image reference 9e08e25fe5dd

telemetry.fabric_outbox  2094 total / 2094 confirmed
gateway evidence         2094 verifications / 2094 verified
```

The historical unclaimed/pending backlog is therefore **not current** at this checkpoint. The normal ULC-01 writer path is draining eligible work successfully.

The HA takeover test was deliberately **not** started. Runtime image parity is not proven: ULC-01 is on the newer fairness-20260915 production image while ULC-02 reports a different image reference/ID. The tracked fairness build artifact records image ID sha256:1641377ae3ed13fbfcbbba663117b4df1cf31bf76087bdc8334390533b1277a9; ULC-02's live 9e08e25fe5dd does not establish parity with it. A disabled standby self-test proves canonicalization/startup behavior, not generation-aware external-side-effect fencing.

Current hard stop: keep ULC-02 FABRIC_ADAPTER_ENABLED=false. Before step 6 below, load and verify the same accepted current adapter build on ULC-02 while it remains disabled, prove the expected distinct FABRIC_ADAPTER_WORKER_ID, re-verify migration 003_fabric_adapter_ha_fencing.sql on the live database, and then execute the controlled failover gate. Never use outbox 1960 for that test.

Production commissioning order

1. Verify ULC-01 is healthy and enabled, ULC-02 is healthy but disabled, and no current critical Fabric errors exist.
2. Apply migration 003_fabric_adapter_ha_fencing.sql once through the current PostgreSQL primary path; verify column/default/check/comment/grant.
3. Load the locked archive on ULC-01 and roll only the Fabric adapter to the generation-aware image. Verify image ID, binary label, enablement, restart count, DB access, OpenBao access, and Fabric path.
4. Load the exact same archive on ULC-02 while keeping FABRIC_ADAPTER_ENABLED=false; verify exact image ID parity and standby self-test/preflight.
5. Verify ULC-01 and ULC-02 use distinct stable FABRIC_ADAPTER_WORKER_ID values.
6. Enable ULC-02 and verify two workers can poll safely without simultaneous ownership of the same live claim.
7. Perform one controlled HA acceptance: stop only the ULC-01 adapter, allow the lease boundary to pass as needed, and prove ULC-02 processes a fresh eligible verified exact-payload row. Never use outbox 1960.
8. Require authoritative Fabric commit, source-bound Query match, and Verify MATCH for the representative takeover row.
9. Restore ULC-01 and verify both workers are healthy, no duplicate/conflicting submission exists, and generation fencing remains authoritative.

10. Leave both workers running only after the preceding gate passes; otherwise leave ULC-02 disabled and retain ULC-01 as sole writer.

Rollback

- If ULC-01 fails after the image rollout, restore its previous known-good image and keep ULC-02 disabled.
- Migration 003 is additive and may remain installed during rollback.
- Never enable the older ULC-02 worker against a dual-worker topology after migration merely because the column exists; both active workers must enforce generation predicates.
- Preserve durable Fabric transaction fields and reconcile unknown submissions before any new Create.

Canonical source: [deployment/technology-guides/15-hyperledger-fabric-adapter.md](#). For any mutable production-status statement, check the dated baseline and verified live state.